

[Description](#)[Package Content](#)[Wiki](#)[WiFi](#), [Bluetooth](#), [ESP32-C6](#), [LCD \(SPI/QSPI\)](#)

ESP32-C6 1.9inch Display Development Board, 170×320 Pixels, 262K Color, 160MHz Running Frequency RISC-V Single-core Processor, WiFi 6 & Bluetooth, Optional For Touch Function, ESP32 With Display

~~\$15.99~~ **\$19.99**

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With Touch Function

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Related Products:



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ESP32-C6 1.9inch Display Development Board, 170×320 Pixels, 262K Color, 160MHz Running Frequency RISC-V Single-Core Processor, WiFi 6 & Bluetooth, Supports 90° Hardware Rotation, Optional For Touch Function, ESP32 With Display

ESP32-C6 1.9inch LCD Display Development Board

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Description

Package Content

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170 × 320, Embedded with ST7789V2 Display Driver, Supports 90° Hardware Rotation





RISC-V 32-bit Processor



2.4 GHz Wi-Fi 6



BLE 5



Onboard Antenna



1.9"



160° Viewing Angle



170×320 Resolution



262K Color



QMI8658 6-Axis IMU



TF Card Slot



Lithium Battery Header



Pico Header Compatibility

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Package Content

Wiki

Version Options

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[Description](#)[Package Content](#)[Wiki](#)**ESP32-C6-LCD-1.9**

without touch control

**ESP32-C6-Touch-LCD-1.9**

with touch control

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Features

This product adopts **ESP32-C6FH8** chip with 2.4GHz Wi-Fi 6 and Bluetooth BLE 5 support, integrates 8MB Flash. Onboard 1.9inch LCD display which can smoothly run GUI programs such as LVGL. Onboard QMI8658 6-axis IMU, TF card slot, Lithium battery header, etc. Compatible with Pico header, offering strong compatibility and expandability. It is suitable for the quick development of the HMI and other ESP32-C6 applications.

- Equipped with a high-performance 32-bit RISC-V processor with clock speed up to 160 MHz
- Integrated WiFi 6, Bluetooth 5 and IEEE 802.15.4 (Zigbee 3.0 and Thread) wireless communication, with superior RF performance
- Built-in 512KB HP SRAM, 16KB LP SRAM, and 320KB ROM, integrates 8MB Flash

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Description	Package Content	WiFi
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Onboard 1.9inch LCD display, 170 × 320

resolution, 262K color, delivering clear and vivid color image display

- Onboard QMI8658 6-axis IMU (3-axis accelerometer and 3-axis gyroscope) for detecting motion gestures, counting steps, etc.
- Onboard USB Type-C port for power supply, program downloading, and debugging, more convenient for development use
- Onboard 3.7V MX1.25 Lithium battery recharge/discharge header
- Onboard TF card slot for external TF card storage of pictures or files
- Compatible with Pico header, onboard multiple peripheral interfaces, offering strong compatibility and expandability

LCD Parameters

DISPLAY PANEL	LCD	DISPLAY SIZE	1.9 inch	Chat
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Description	Package Content	Wiki	DISPLAY COLORS	
RESOLUTION		170×320		262K
BRIGHTNESS		500 cd/m ²	CONTRAST RATIO	900:1
COMMUNICATION INTERFACE		SPI	DRIVER IC	ST7789V2
TOUCH		Supported (Touch Version Only)	TOUCH IC	CST816 (Touch Version Only)

Application Scenarios

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Human-machine Interface

The Human-machine Interface (also known as the user interface) is the medium of interaction and information exchange between the system and the user, it realizes the transformation between the internal form of information and the form acceptable to human beings.



LVGL GUI Development

LVGL is a free, open-source graphics library that provides everything you need to create embedded GUI with the easy-to-use graphical elements, beautiful visual effects and low memory requirement.

Supports ESP-IDF, Arduino IDE

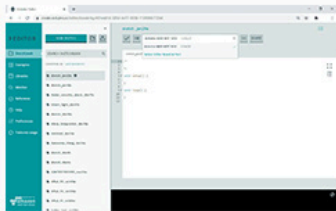
Comprehensive SDK, Dev Resources, Tutorials To Help You Easily Get Started

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ESP-IDF

With free open source development tools, supports IDEs such as VS Code and Eclipse easier for developers to use.



Arduino IDE

Arduino IDE is an open source electronic prototyping platform, convenient and flexible, easy to get started.

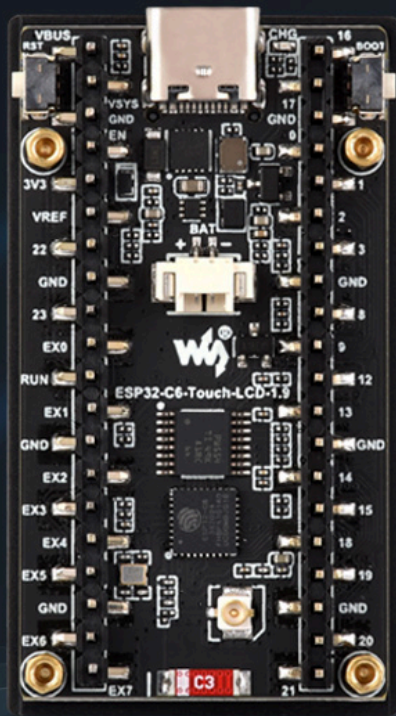
Wi-Fi 6 And BLE 5 Support

Supports WiFi 6, Bluetooth 5 and IEEE 802.15.4 (Zigbee 3.0 and Thread)

Its Bluetooth Low Energy Subsystem Supports Bluetooth 5 (LE) And Bluetooth Mesh

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2.4GHz Wi-Fi 6


Bluetooth 5 (LE)

Interface Definition

Compatible With Pico Header, Can Be Used With Some Raspberry Pi Pico HATs, Offering Strong Compatibility And Expandability

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[illegible]

Description

Package Content

Wiki



Display Control		
LCD pin	ESP32-C6 pin	Description
LCD_DIN	GPIO4	LCD data pin
LCD_CLK	GPIO5	LCD clock pin
LCD_DC	GPIO6	LCD command/ data selection
LCD_CS	GPIO7	LCD chip selection
LCD_RST	GPIO14	LCD reset pin
LCD_BL	GPIO15	LCD backlight

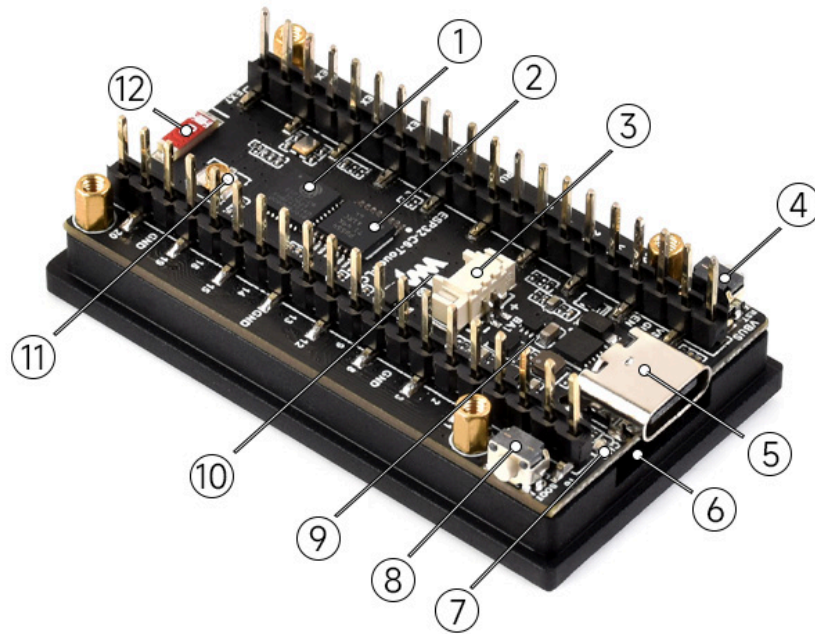


Touch Control		
LCD pin	ESP32-C6 pin	Description
TP_SCL	GPIO8	TP I2C clock pin
TP_SDA	GPIO18	TP I2C data pin
TP_RST	GPIO21	TP reset pin (NC)
TP_INT	GPIO22	TP interrupt output pin (NC)

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What's On Board

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[Description](#) [Package Content](#) [Wiki](#)

1. **ESP32-C6FH8**

integrates 8MB Flash, up to 160MHz operating frequency

2. **TCA9554**

8-ch GPIO expander chip, enhances IO flexibility

3. **MX1.25 battery header**

MX1.25 2PIN connector, for 3.7V Lithium battery, supports charging and discharging

4. **RESET button**

5. **USB Type-C port**

for power input and data transmission

6. **TF card slot (on the back side)**

Supports TF card for extended storage

7. **Charge indicator**

Lithium battery charge indicator, lights up when charging, off when fully charged

8. **BOOT button**

for downloading program or entering boot mode

9. **WS2812 RGB LED (for non-touch version only)**

for colorful lighting effects, on the back side of the PCB

10. **QMI8658 sensor (on the back side)**

6-axis IMU includes a 3-axis gyroscope and a 3-axis accelerometer

11. **IPEX1 connector (for touch version only)**

Switching to use external antenna via resoldering an onboard resistor

12. **Onboard ceramic antenna**

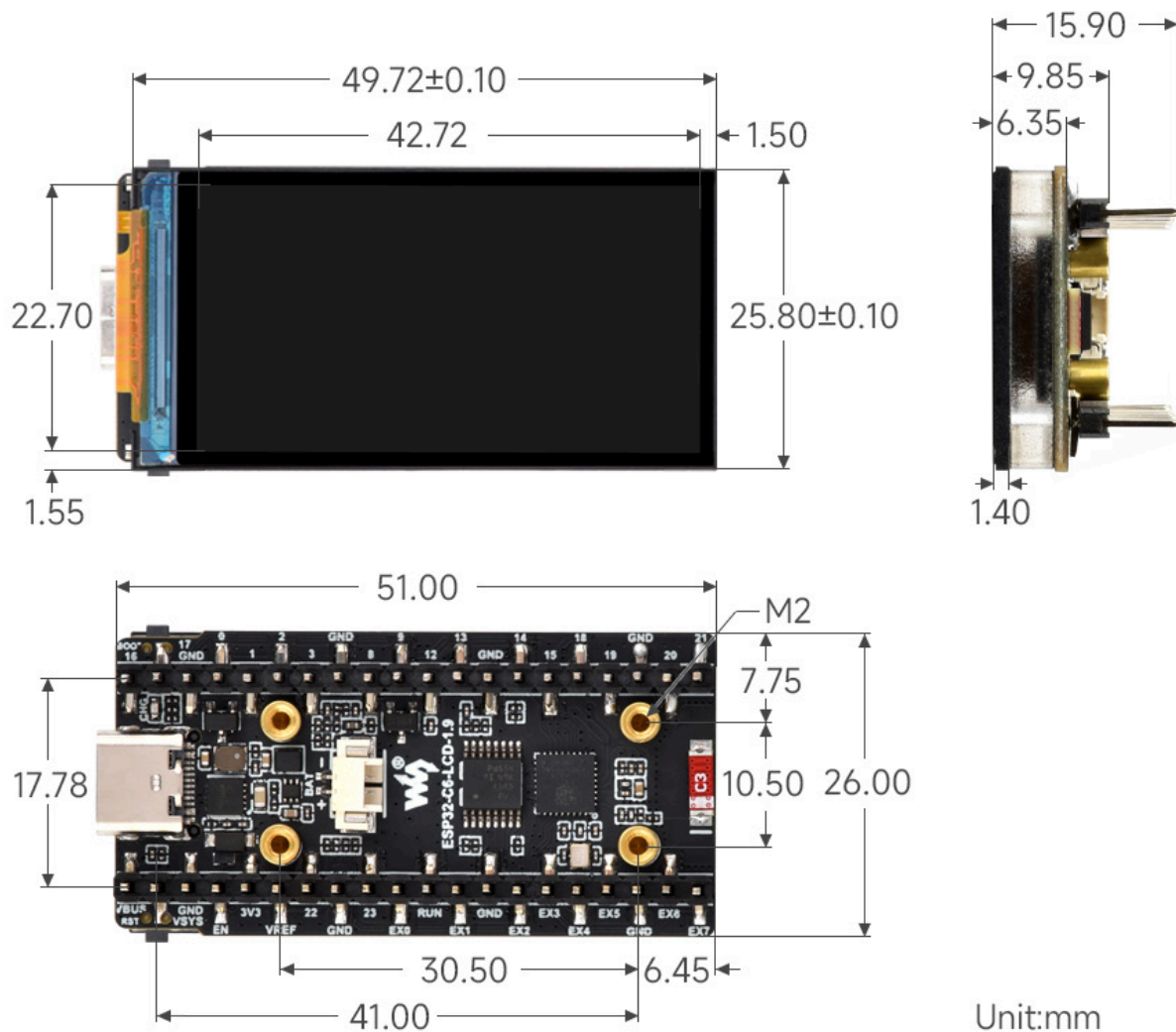
Onboard antenna design for space saving and signal stability

Outline Dimensions

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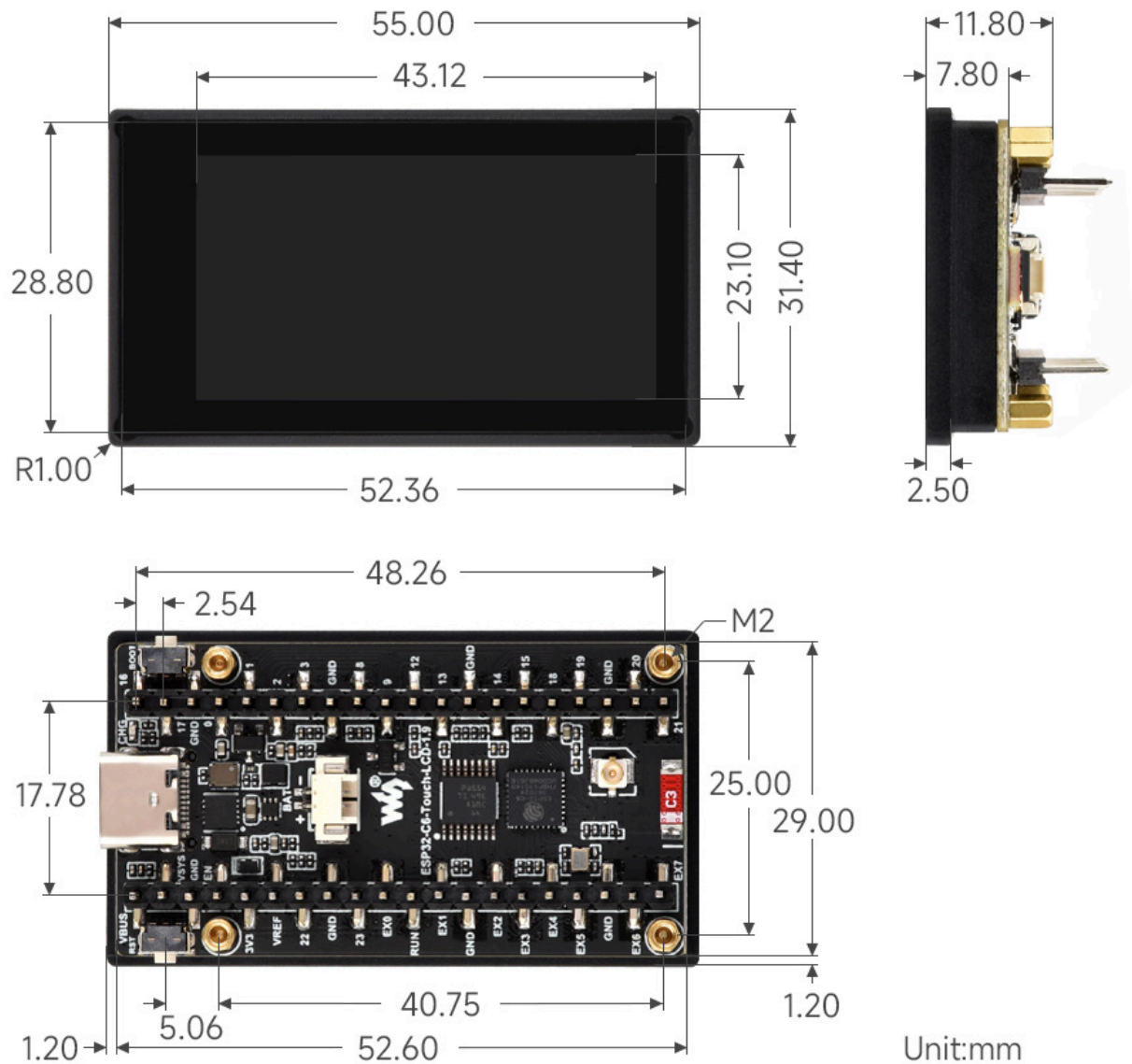
Without Touch Version



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With Touch Version



Resources & Services



WAVESHARE
ORIGINAL ✓



PROFESSIONAL
TECH SUPPORT



RICH RESOURCES
MANUAL & DEMO

Wiki: www.waveshare.com/wiki/ESP32-C6-LCD-1.9

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Description

Package Content

Wiki

ESP32 LCD Selection

MODEL	CORE CONFIGURATION			DISPLAY				ONBOARD	
	CPU	PSRAM	FLASH	PIXELS	INTERFACE	DISPLAY	TOUCH	IMU	R1
AMOLED DISPLAY									
 ESP32-S3-Touch-AMOLED-2.41	ESP32-S3R8	8MB	16MB	600 × 450	QSPI	RM690B0	FT6336	√	√
 ESP32-S3-Touch-AMOLED-2.41-B	ESP32-S3R8	8MB	16MB	600 × 450	QSPI	RM690B0	FT6336	√	√
 ESP32-C6-Touch-AMOLED-2.16	ESP32-C6	-	16MB	480 × 480	QSPI	CO5300	CST9220	√	√
 ESP32-S3-Touch-AMOLED-2.16	ESP32-S3R8	8MB	16MB	480 × 480	QSPI	CO5300	CST9220	√	√

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Description	Package Content	Wiki						
 ESP32-S3-Touch-AMOLED-2.06	ESP32-S3R8 8MB	32MB	410 × 502	QSPI	CO5300	FT3168	√	√
 ESP32-C6-Touch-AMOLED-2.06	ESP32-C6 -	16MB	410 × 502	QSPI	CO5300	FT3168	√	√
 ESP32-S3-Touch-AMOLED-1.91	ESP32-S3R8 8MB	16MB	240 × 536	QSPI	RM67162	FT3168	√	-
 ESP32-S3-AMOLED-1.91	ESP32-S3R8 8MB	16MB	240 × 536	QSPI	RM67162	-	√	-
 ESP32-C6-Touch-AMOLED-1.8	ESP32-C6 -	16MB	368 × 448	QSPI	SH8601	FT3168 / FT6146	√	√
 ESP32-S3-Touch-AMOLED-1.8	ESP32-S3R8 8MB	16MB	368 × 448	QSPI	SH8601	FT3168 / FT6146	√	√

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Description	Package	Content	Wiki							
 ESP32-S3-Touch-AMOLED-1.75	ESP32-S3R8	8MB	16MB	466 × 466	QSPI	CO5300	CST9217	√	√	
 ESP32-S3-Touch-AMOLED-1.75-B	ESP32-S3R8	8MB	16MB	466 × 466	QSPI	CO5300	CST9217	√	√	
 ESP32-S3-Touch-AMOLED-1.75C	ESP32-S3R8	8MB	32MB	466 × 466	QSPI	CO5300	CST9217	√	-	
 ESP32-C6-Touch-AMOLED-1.64	ESP32-C6	-	16MB	280 × 456	QSPI	CO5300	FT6146	√	-	
 ESP32-S3-Touch-AMOLED-1.64	ESP32-S3R8	8MB	16MB	280 × 456	QSPI	CO5300	FT3168	√	-	
 ESP32-C6-Touch-AMOLED-1.43	ESP32-C6	-	16MB	466 × 466	QSPI	CO5300	FT6146	√	√	

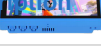
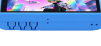
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Description	Package Content	Wiki						
 ESP32-C6-Touch-AMOLED-1.43-B	ESP32-C6 -	16MB	466 × 466	QSPI	CO5300	FT6146	√	√
 ESP32-S3-Touch-AMOLED-1.43	ESP32-S3R8 8MB	16MB	466 × 466	QSPI	SH8601 /CO5300	FT3168 /FT6146	√	√
 ESP32-S3-Touch-AMOLED-1.43-B	ESP32-S3R8 8MB	16MB	466 × 466	QSPI	SH8601 /CO5300	FT3168 /FT6146	√	√
 ESP32-S3-Touch-AMOLED-1.43C	ESP32-S3 8MB	8MB	466 × 466	QSPI	CO5300	CST820	-	-
 ESP32-C6-Touch-AMOLED-1.32	ESP32-C6 -	16MB	466 × 466	QSPI	CO5300	CST820	-	-
 ESP32-C6-Touch-AMOLED-1.32	ESP32-C6 -	16MB	466 × 466	QSPI	CO5300	CST820	-	-

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Description	Package Content	Wiki						
 ESP32-S3-Touch-AMOLED-1.32	ESP32-S3-PICO-1-N8R8 8MB	8MB	466 × 466	QSPI	CO5300	CST820	-	-
LCD DISPLAY								
 ESP32-Touch-LCD-3.5	ESP32-D0WDR2-V3 2MB	16MB	320 × 480	SPI	ST7796	FT6336	-	√
 ESP32-S3-Touch-LCD-3.5	ESP32-S3R8 8MB	16MB	320 × 480	SPI	ST7796	FT6336	√	√
 ESP32-S3-Touch-LCD-3.5-C	ESP32-S3R8 8MB	16MB	320 × 480	SPI	ST7796	FT6336	√	√
 ESP32-S3-Touch-LCD-3.5B	ESP32-S3R8 8MB	16MB	320 × 480	QSPI	AXS15231B	AXS15231B	√	√
 ESP32-S3-Touch-LCD-3.5B-C	ESP32-S3R8 8MB	16MB	320 × 480	QSPI	AXS15231B	AXS15231B	√	√

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	Description	Package Content	Wiki								
	 ESP32-S3-Touch-LCD-3.49	ESP32-S3R8	8MB	16MB	172 × 640	QSPI	AXS15231B	AXS15231B	√	√	
	 ESP32-S3-Touch-LCD-3.49B	ESP32-S3R8	8MB	16MB	172 × 640	QSPI	AXS15231B	AXS15231B	√	√	
	 ESP32-S3-LCD-3.16	ESP32-S3R8	8MB	16MB	320 × 820	RGB	ST7701	-	√	√	
	 ESP32-S3-Touch-LCD-2.8	ESP32-S3R8	8MB	16MB	240 × 320	SPI	ST7789	CST328	√	√	
	 ESP32-S3-LCD-2.8	ESP32-S3R8	8MB	16MB	240 × 320	SPI	ST7789	-	√	√	
	 ESP32-S3-Touch-LCD-2.8B	ESP32-S3R8	8MB	16MB	480 × 640	RGB	ST7701	GT911	√	√	
	 ESP32-S3-LCD-2.8B	ESP32-S3R8	8MB	16MB	480 × 640	RGB	ST7701	-	√	√	

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Description	Package Content	Wiki							
 ESP32-S3-Touch-LCD-2.8C	ESP32-S3R8	8MB	16MB	480 × 480	RGB	ST7701	GT911	√	√
 ESP32-S3-LCD-2.8C	ESP32-S3R8	8MB	16MB	480 × 480	RGB	ST7701	-	√	√
 ESP32-S3-Touch-LCD-2.1	ESP32-S3R8	8MB	16MB	480 × 480	RGB	ST7701	CST820	√	√
 ESP32-S3-Touch-LCD-2.1B	ESP32-S3R8	8MB	16MB	480 × 480	RGB	ST7701	CST820	√	√
 ESP32-S3-Touch-LCD-2	ESP32-S3R8	8MB	16MB	240 × 320	SPI	ST7789T3	CST816	√	-
 ESP32-S3-LCD-2	ESP32-S3R8	8MB	16MB	240 × 320	SPI	ST7789T3	-	√	-
 ESP32-S3-Touch-LCD-1.9	ESP32-S3R8	8MB	16MB	170 × 320	SPI	ST7789	CST816	√	-
 ESP32-S3-LCD-1.9	ESP32-S3R8	8MB	16MB	170 × 320	SPI	ST7789	-	√	-

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	 Description	Package Content		Wiki							
	ESP32-C6-Touch-LCD-1.9	ESP32-C6FH8	-	8MB	170 × 320	SPI	ST7789	CST816	√	-	
	 ESP32-C6-LCD-1.9	ESP32-C6FH8	-	8MB	170 × 320	SPI	ST7789	-	√	-	
	 ESP32-S3-Touch-LCD-1.85C	ESP32-S3R8	8MB	16MB	360 × 360	QSPI	ST77916	CST816	-	√	
	 ESP32-S3-Touch-LCD-1.85C-BOX	ESP32-S3R8	8MB	16MB	360 × 360	QSPI	ST77916	CST816	-	√	
	 ESP32-S3-Touch-LCD-1.85B	ESP32-S3R8	8MB	16MB	360 × 360	QSPI	ST77916	CST816	√	√	
	 ESP32-S3-Touch-LCD-1.85	ESP32-S3R8	8MB	16MB	360 × 360	QSPI	ST77916	CST816	√	√	
	 ESP32-S3-LCD-1.85	ESP32-S3R8	8MB	16MB	360 × 360	QSPI	ST77916	-	√	√	Chat

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Image	Description	Package	Content	Wiki							
	ESP32-S3-Touch-LCD-1.83	ESP32-S3R8	8MB	16MB	240 × 284	SPI	ST7789P	CST816D	√	√	
	ESP32-C6-Touch-LCD-1.83	ESP32-C6	-	16MB	240 × 284	SPI	ST7789P	CST816D	√	√	
	ESP32-C6-Touch-LCD-1.69	ESP32-C6	-	16MB	240 × 280	SPI	ST7789V2	CST816T	√	√	
	ESP32-C6-LCD-1.69	ESP32-C6	-	16MB	240 × 280	SPI	ST7789V2	-	√	√	
	ESP32-S3-Touch-LCD-1.69	ESP32-S3R8	8MB	16MB	240 × 280	SPI	ST7789	CST816	√	√	
	ESP32-S3-LCD-1.69	ESP32-S3R8	8MB	16MB	240 × 280	SPI	ST7789	-	√	√	
	ESP32-C6-LCD-1.54	ESP32-C6	-	16MB	240 × 240	SPI	ST7789	-	√	-	

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Description	Package Content	Wiki						
 ESP32-C6-Touch-LCD-1.54	ESP32-C6 -	16MB	240 × 240	SPI	ST7789	CST816	√	-
 ESP32-S3-LCD-1.54	ESP32-S3R8 8MB	16MB	240 × 240	SPI	ST7789	-	√	-
 ESP32-S3-Touch-LCD-1.54	ESP32-S3R8 8MB	16MB	240 × 240	SPI	ST7789	CST816	√	-
 ESP32-S3-Touch-LCD-1.47	ESP32-S3R8 8MB	16MB	172 × 320	SPI	JD9853	AXS5106L	-	-
 ESP32-S3-LCD-1.47	ESP32-S3R8 8MB	16MB	172 × 320	SPI	ST7789	-	-	-
 ESP32-S3-LCD-1.47B	ESP32-S3R8 8MB	16MB	172 × 320	SPI	ST7789	-	√	-
 ESP32-C6-Touch-LCD-1.47	ESP32-C6FH8 -	8MB	172 × 320	SPI	JD9853	AXS5106L	√	-

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Descrip		Package Content	Wiki							
ESP32-C6-LCD-1.47		ESP32-C6FH8	-	8MB	172 × 320	SPI	ST7789	-	-	-
ESP32-C3-LCD-1.47		ESP32-C3FH4	-	4MB	172 × 320	SPI	ST7789	-	-	-
ESP32-S3-Touch-LCD-1.46B		ESP32-S3R8	8MB	16MB	412 × 412	QSPI	SPD2010	SPD2010	√	√
ESP32-S3-Touch-LCD-1.46		ESP32-S3R8	8MB	16MB	412 × 412	QSPI	SPD2010	SPD2010	√	√
ESP32-C6-LCD-1.3		ESP32-C6FH4	-	4MB	240 × 240	SPI	ST7789	-	-	-
ESP32-C6-LCD-1.3-M		ESP32-C6FH4	-	4MB	240 × 240	SPI	ST7789	-	-	-
ESP32-S3-LCD-1.3		ESP32-S3R8	8MB	16MB	240 × 240	SPI	ST7789	-	√	-

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Description	Package Content	Wiki						
 ESP32-S3-LCD-1.3-B	ESP32-S3R8	8MB	16MB	240 × 240	SPI	ST7789	-	√
 ESP32-S3-LCD-1.3-C	ESP32-S3R8	8MB	16MB	240 × 240	SPI	ST7789	-	√
 ESP32-S3-Touch-LCD-1.28	ESP32-S3R2	2MB	16MB	240 × 240	SPI	GC9A01A	CST816	√
 ESP32-S3-LCD-1.28	ESP32-S3R2	2MB	16MB	240 × 240	SPI	GC9A01A	-	√
 ESP32-C3-LCD-0.71	ESP32-C3FH4	-	4MB	160 × 160	SPI	GC9D01	-	-
 ESP32-S3-LCD-0.85	ESP32-S3	8MB	8MB	128 × 128	SPI	GC9107	-	-
 ESP32-C6-LCD-0.85	ESP32-C6	-	8MB	128 × 128	SPI	ST7735	-	-
 ESP32-C6-GEEK	ESP32-C6	-	16MB	240 × 135	SPI	ST7789P3	-	- Chat

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		ESP32-	2MB	16MB	240 × 135	SPI	ST7789P3	-	-	-	
	ESP32-S3-GEEK	S3R2									

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